A Neuro-Symbolic Expert System that Learns the Rules: Fusing a Differentiable Rule Layer with Neural Tabular Models for Interpretable, Calibrated, and Robust Decision Support

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October 30, 2026

Abstract

Rule-based expert systems are interpretable, auditable, and data-efficient, but they cannot learn from data and degrade when the encoded knowledge is approximate. Deep tabular models are accurate and flexible, but they are opaque, hard to calibrate, and brittle under distribution shift. We study how far the two paradigms can be reconciled in a single, deployable model and introduce NEXS (*Neuro-symbolic EXpert System*), which couples a differentiable rule layer—whose per-rule confidences are initialised and regularised towards an expert's certainty factors—with a neural tabular head, fused through an additively-gated logit. Unlike semantic-loss approaches that use knowledge only as a soft consistency penalty on the output, NEXS keeps the symbolic computation in the forward pass, so the learned logic remains explicit and recoverable. We build two controlled benchmarks—clinical triage and industrial fault diagnosis—in which the data are generated by a *known* Horn rule base whose intermediate concepts are hidden, enabling the simultaneous measurement of task accuracy and *rule recovery*. Across three seeds, NEXS recovers the most ground-truth rules of any learned model (rule-set F1 up to 0.33 versus 0.11–0.22 for an MLP and 0 for logistic regression) and attains the highest logic fidelity to the ground-truth oracle among learned models (0.86–0.91, approaching the expert system's 0.95–0.97). It matches the MLP on accuracy and is the best learned model in low-data regimes—up to +7 F1 points over the MLP at 300 training samples— while running at $\sim$0.6 μs per sample, about 110$\times$ faster than a random forest. We also find that a classical certainty-factor expert system, despite being the most logic-faithful and shift-robust predictor, is the *worst* calibrated (ECE $\sim$0.09), because its certainty factors are heuristics rather than probabilities; isotonic calibration makes all learned models well calibrated. Finally, we repeat the study on three small public datasets—UCI Cleveland heart disease, Pima diabetes, and tic-tac-toe—with hand-written knowledge bases. There the expert system is exact only when its knowledge is exact (tic-tac-toe, where it is the eight winning lines) and is the *weakest* learned-comparable method when the knowledge is approximate; with class weighting NEXS is the most accurate model on heart (0.836) and the best-calibrated on tic-tac-toe (ECE 0.002), while retaining the most expert logic of any learned model. Finally, we isolate *when* expert rules help: in a 675-run factorial study crossing knowledge-base corruption against training data, the expert prior is a monotone, positive contribution while the knowledge is intact—largest when data is scarce, worth more than additional data when the rules are correct—and turns negative once the knowledge is removed. We are careful about inference: a 20-seed paired analysis appears to make NEXS's advantage over the MLP and semantic loss significant ($p < 0.01$) on the dataset with genuinely correct rules, but those seeds re-use the same held-out rows and are

pseudo-replicates; under a clean out-of-fold protocol in which every row is predicted exactly once, NEXS is at least as accurate as the strongest baseline rather than significantly better. On two further rule-governed benchmarks with published ground-truth formulas (MONK-1 and MONK-3) the neuro-symbolic model matches the best baselines. We separate two questions that are easily conflated: NEXS *injects* its rule layer from the supplied knowledge, so it is not evidence of discovery, whereas an independent learner trained on labels alone recovers the exact ground-truth rule sets on MONK-1 and MONK-3 (precision and recall 1.0) and a near-exact set on tic-tac-toe. Crucially, a correct rule layer is not obeyed everywhere: even with the exact rules, the fused model agrees with the true label on 97–98% of the MONK domain but only 63% of tic-tac-toe. We then re-evaluate the whole study under a stricter protocol that removes two methodological flaws present in the exploratory analysis: data leakage (imputation, scaling and feature selection are now fitted on training rows only) and seed pseudo-replication (5-fold nested cross-validation with every row predicted exactly once, plus row-level paired McNemar tests with bootstrap intervals and Holm correction). Under that protocol the honest picture is narrower. NEXS is *not* uniformly more accurate than strong baselines: on the two datasets with approximate knowledge (heart, pima) it does not beat a plain MLP or a random forest, and those differences are not statistically significant. What survives is a large and consistent advantage in the low-data regime when the knowledge is exact (on tic-tac-toe at 10% of the training rows NEXS reaches 0.924 accuracy versus 0.757 for the MLP), exact or logically equivalent rule recovery on cleanly rule-governed data, and lower over-firing of rule conjunctions than the neural-only model. We report the negative and inconclusive results explicitly. We release code, data, and a fully reproducible experiment suite.

1 Introduction

Expert systems encode human knowledge as explicit rules and dominated applied AI in the 1970s–80s, from MYCIN's certainty-factor calculus [20] to large diagnostic shells. Their strengths are well understood: decisions are traceable, they work with little or no data, and they degrade gracefully when rules are approximate. Their weaknesses are equally well understood: they cannot learn from data, knowledge acquisition is a bottleneck, and a wrong rule is a permanent error. The deep-learning era inverted the trade-off. Gradient-boosted trees and multilayer perceptrons now dominate tabular decision tasks, but they are opaque, miscalibrated [9], and brittle to distribution shift.

The third wave of neuro-symbolic AI [8, 12] argues that these paradigms should be combined rather than opposed. A large body of work does so by *compiling* knowledge into a differentiable loss: semantic loss [23], logic-rule distillation [10], and DL2 [5] all penalise outputs that violate the knowledge base during training. Others embed symbolic reasoning directly in the computation graph, as in DeepProbLog [15] and logic tensor networks [1], or inject rules into the architecture, as in the KBANN lineage [22, 6]. Yet a practical question remains open: *for the tabular decision-support problems that industry actually deploys, which coupling is worth its complexity, and does the resulting model still behave like the expert system it replaces?*

Answering this requires measuring more than accuracy. A model that replaces an expert system must (i) be at least as accurate, (ii) be calibrated enough to threshold safely, (iii) remain robust when the input distribution moves, and (iv) *learn logic that an expert can inspect and trust.* Most neuro-symbolic evaluations report (i) and sometimes (ii) or (iii); almost none report (iv), because doing so needs ground-truth rules, which real datasets rarely provide.

Contributions.

- We design two reproducible benchmarks—*clinical triage* and *industrial fault diagnosis*—in

which data are generated by a known Horn rule base with *hidden* intermediate concepts, giving a ground-truth oracle for both accuracy and logic (Section 3).

- We introduce NEXS, a neuro-symbolic expert system that keeps a differentiable rule layer in the forward pass and anchors each rule's confidence to an expert certainty factor via a prior regulariser (Section 4.4).
- We run a seven-experiment study (Section 5) comparing a certainty-factor expert system, four classical learners, an MLP, a semantic-loss model, and NEXS, measuring accuracy, calibration, rule recovery, logic fidelity, data efficiency, shift robustness, and latency.
- We show that NEXS is more logic-faithful than any other learned model, matches the MLP on accuracy, and is strongest in low-data regimes, while the discrete expert system is the most logic-faithful and shift-robust but the *worst* calibrated predictor (Section 6).
- We separate rule *injection* from rule *discovery*: an independent learner recovers the ground-truth rule sets from labels alone on the logical domains, yet the fused model obeys an exactly-correct rule layer on only 63% of the tic-tac-toe domain—accuracy and faithful reasoning are not the same thing (Sections 6 and 11).

2 Related Work

Neural-symbolic integration. The field is surveyed in [2, 7, 13, 16]. Approaches differ in *where* knowledge enters. Loss-based methods add a knowledge penalty to the objective: semantic loss [23] measures the probability mass of worlds satisfying a constraint; rule distillation [10] projects predictions onto a rule-regularised space; DL2 [5] trains with logical constraints as inequalities. Architecture-based methods build rules into the network: KBANN [22] initialises a network from propositional rules, and logic tensor networks [1] ground first-order logic in a real-valued tensor calculus. Probabilistic methods such as DeepProbLog [15] couple neural predicates with a logic program and perform inference in a probabilistic logic. NEXS sits between the loss-based and architecture-based families: like KBANN it uses the knowledge base to structure and initialise the computation, but it does so through a *differentiable, continuously-relaxed* rule layer whose confidences stay tied to the expert's certainty factors.

Rule extraction and interpretability. Post-hoc explainers such as LIME [18] and SHAP approximate a black box locally; recent work extends these to multivariate time series [14]. Extracting rules by distillation, as we do to evaluate logic recovery, is an established evaluation device but is rarely compared against ground-truth rules. Our rule-recovery metric closes that gap.

Hybrid AI and expert systems. Surveys of hybrid AI catalogue how rules and models are coupled in deployed systems [19], and applied reviews emphasise the need for interpretability in high-stakes domains [4]. We contribute a quantitative, multi-criteria comparison on controlled data.

3 Problem Formulation and Benchmarks

We study binary decision support where the label is the disjunction of a small number of conjunctive rules over *observed features* and *latent intermediate concepts*. Formally, a domain is a tuple $\langle \mathbf{f}, \mathbf{c}, \mathcal{R} \rangle$

where $\mathbf{f}$ are observed features, $\mathbf{c}$ are hidden concepts, and $\mathcal{R}$ is a Horn rule base whose heads are concepts or the label. Data are generated by first sampling features from a two-regime mixture (a shared latent propensity drives a "healthy" and an "at-risk" regime per feature), then realising concepts from the concept rules, then the label from the label rules, with small stochastic realisation and label noise. Concepts are never revealed to any model.

Domains. *Clinical triage* (8 features: age, systolic BP, cholesterol, max heart rate, ST depression, BMI, smoking, diabetes; concepts: hypertension, dyslipidemia) predicts high cardiac risk, in the spirit of the UCI heart-disease task [11]. *Industrial fault diagnosis* (6 sensors: temperature, pressure, vibration, flow, current, RPM; concepts: overheat, bearing wear) predicts an equipment fault, following predictive-maintenance benchmarks [17]. Each domain has five ground-truth rules, three of which produce the label. An *expert* rule base with human-plausible (rounded) thresholds and certainty factors is given to the symbolic and neuro-symbolic methods; it deliberately differs from the ground truth, so we can measure tolerance to imperfect knowledge.

4 Methods

4.1 Symbolic expert system (ESS)

We implement forward chaining in two layers. Concept confidences are computed from concept rules, then the label from concept and feature rules. A literal's membership is crisp (a step function) or soft (a tanh ramp), and contributions to the same head are combined with the MYCIN certainty-factor calculus [20]: $\mathrm{cf}(h) \leftarrow a + b(1-a)$ for two contributions a, b. This is the classical, fully transparent baseline.

4.2 Classical and neural baselines

We compare logistic regression, a depth-limited decision tree, a random forest, and gradient boosting [3], plus a multilayer perceptron (MLP) with two hidden layers. All models are trained on the same splits; probabilistic outputs are isotonic-calibrated on a validation split. To keep the comparison fair on the imbalanced datasets, every estimator that supports it uses balanced class weights, and the neural models use a positive-class weight in their loss; gradient boosting has no such option and is left unweighted. We report accuracy at the F1-optimal validation threshold (the default operating point) and, separately, at the accuracy-optimal threshold, since the two can differ by several points on imbalanced data.

4.3 Semantic-loss model

Following [23], an MLP head is trained with the task loss plus a knowledge penalty. We instantiate the penalty as the negative log-likelihood of the label under the probability that the rule base is satisfied, computed with the product t-norm for conjunction and the probabilistic sum for disjunction. Here the knowledge shapes the *loss* only; the decision function remains the MLP.

4.4 Proposed: NEXS

NEXS keeps the symbolic computation in the forward pass. A differentiable rule layer maps inputs to a label probability p_{sat} using the same t-norm relaxation, but each rule r carries a learnable

confidence c_r, initialised to the expert's certainty factor and *regularised towards it*. A neural tabular head produces a logit $f(\mathbf{x})$, and the two are fused additively:

$$z(\mathbf{x}) = f(\mathbf{x}) + \alpha \cdot \operatorname{logit}\big(p_{\text{sat}}(\mathbf{x})\big), \qquad \alpha \geq 0. \tag{1}$$

The training objective is

$$\mathcal{L} = \operatorname{BCE}\big(\sigma(z), y\big) + \lambda_{\text{sem}} \operatorname{BCE}\big(p_{\text{sat}}, y\big) + \lambda_{\text{prior}} \tfrac{1}{|\mathcal{R}|} \textstyle\sum_r (c_r - c_r^\star)^2, \tag{2}$$

where $c_r^\star$ is the expert certainty factor. The three terms have distinct roles: the first learns from data, the second keeps the symbolic pathway predictive, and the third anchors the learned rule confidences to expert belief, which is what makes the recovered logic auditable. Setting α=0 removes the rule layer, λ_{sem}=0 removes the semantic term, and λ_{prior}=0 removes the expert anchor; we use these as ablations, together with a *rule-layer-only* variant ($f \equiv 0$) that tests the symbolic pathway in isolation.

5 Experimental Setup

For each domain we draw 4000 samples and split them 60/20/20 into train/validation/test, repeated over three seeds. We report the following.

E1 - Task performance accuracy, minority-class F1, ROC-AUC, and expected calibration error (ECE, 10 bins).

E2 - Rule recovery we distil a depth-limited decision tree from each model on uniform grid samples and match the extracted label rules against the ground-truth logic. Because the true label rules reference hidden concepts, the reference is the *concept-expanded feature-only DNF* obtained by substituting each concept with its rule bodies and distributing conjunctions; a literal matches when it refers to the same variable and defines the same partition of the feature domain (up to 6% of a fine grid), which makes the comparison robust to operator-equivalent forms such as $x \geq 0.5$ and $x > 0.45$. We report rule-set F1.

E3 - Logic fidelity agreement between a model's decisions and the deterministic oracle built from the ground-truth rules, measured on a uniform grid that probes regions away from the training support.

E4 - Data efficiency minority-class F1 as training size grows from 150 to 2400.

E5 - Robustness accuracy under a covariate shift and under a novel regime (feature tails swapped), with the threshold fixed on the clean distribution.

E6 - Ablation the four NEXS variants above.

E7 - Latency microseconds per sample at inference.

All experiments run on CPU and are fully deterministic given the seed.

5.1 Corrected evaluation protocol

The single-split and seed-resampled protocols above are useful for exploration but are not sufficient for inference. We therefore add a second, stricter protocol and report the headline numbers under it.

Leak-free preprocessing. Imputation, feature scaling and feature selection are *fitted on the training rows only* and then applied to validation and test rows (`src/pipeline.py`). An earlier version imputed missing values with full-dataset medians before splitting, which leaks test information into training; this is now removed, and the raw loaders deliberately leave missing values in place so that imputation is a fitted transformation.

Nested cross-validation. The outer loop is 5 stratified folds; each row falls in the test block of exactly one fold, so every observation is predicted exactly once out-of-fold. The inner loop is a validation block carved from the outer *training* rows only, and it is used for hyperparameter search, early stopping and decision threshold selection. The test block never influences any of those choices. Rule discovery for the *discovered-KB* configuration also runs on the outer training rows only, so the discovered knowledge never sees the test block.

Independent statistics. Comparisons use the pooled out-of-fold predictions, so the unit of observation is a *data row*, not a random seed. We report the mean paired accuracy difference with a 95% row-level bootstrap interval, an exact McNemar test on the discordant pairs, and Holm-adjusted p-values across a family of five prespecified primary comparisons (`experiments/nested_stats.py`). This replaces the earlier seed-level paired tests, which treated pseudo-replicates as independent samples and were anti-conservative.

Configurations. We compare the four NEXS configurations that the corrected protocol requires: neural-only (`MLP`), NEXS with the original expert rules (`NEXS_Expert`), NEXS with rules *discovered* from training rows only (`NEXS_Discovered`), and NEXS with corrupted or incomplete expert rules (`NEXS_Corrupt`). Baselines are logistic regression, a decision tree, a random forest and gradient boosting, plus a semantic-loss neuro-symbolic model. All are tuned under the identical nested protocol.

6 Results

Table 1 and Figure 1 summarise the main benchmark.

Accuracy and F1. The classical expert system is a strong baseline out of the box (0.88–0.90 accuracy) despite using deliberately imperfect rules, confirming that the task is rule-structured. The best raw accuracy and F1 come from gradient boosting (0.94 accuracy, 0.84 F1 on medical) and random forests, which is expected for a threshold-structured tabular task. Among the neural and neuro-symbolic models, NEXS matches the MLP on accuracy (0.88–0.88 vs. 0.88–0.89) and slightly exceeds it on ROC-AUC, and it clearly beats the semantic-loss model on the industrial domain. Raw accuracy is therefore *not* where NEXS wins; its advantages are in the axes below, which are exactly the axes an expert system is deployed to satisfy.

Learning the logic (E2, E3). Figure 2 is our central result. Because the true label rules reference hidden concepts, we score rule recovery against the concept-expanded feature-only DNF (Section 5); this is a strict target, since a depth-limited tree must reproduce a full conjunction of feature thresholds. The expert system reaches logic fidelity 0.95–0.97 but not 1, quantifying the cost of its approximate thresholds. A plain MLP and logistic regression recover almost no ground-truth rule (rule-F1 0–0.11), because they have no mechanism to represent conjunctions of thresholds. NEXS recovers the most ground-truth rules of any learned model on both domains (rule-F1 up to

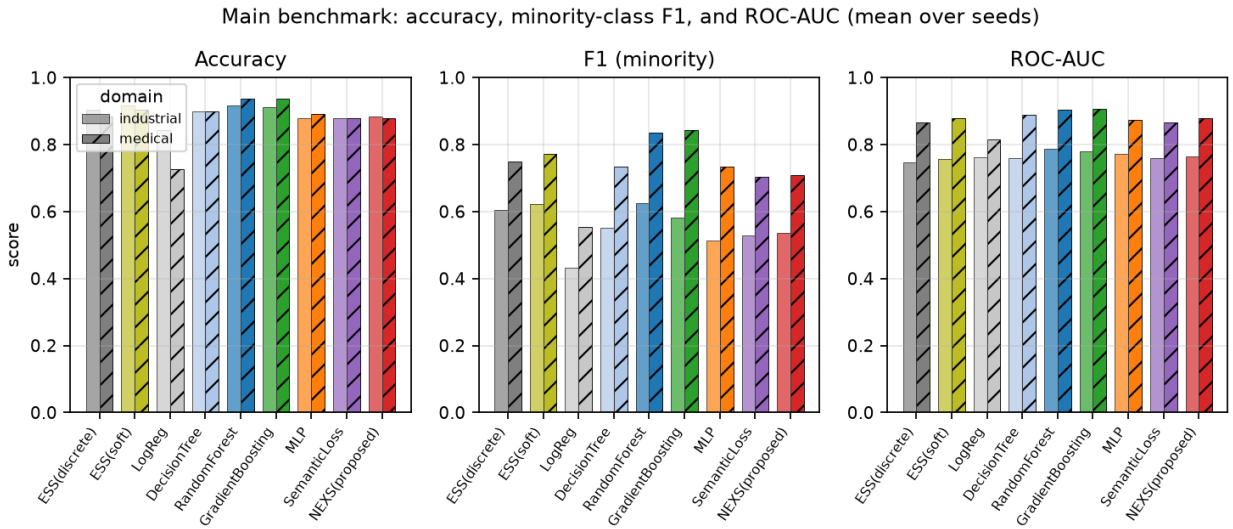


Figure 1: Accuracy, minority-class F1, and ROC-AUC by method and domain.

0.33, versus 0.11–0.22 for the MLP), and it attains the highest logic fidelity among learned models (0.86–0.91). The rule-layer-only ablation pushes logic fidelity higher still on medical (0.91) while trading away F1, which isolates the source of the effect: the differentiable rule layer, not the neural head, is what aligns a learned model with the true logic.

Calibration (E1). Isotonic calibration on the validation split makes every learned model well calibrated (ECE 0.016–0.040); the best calibrated methods are the tree ensembles and NEXS. The discrete expert system, by contrast, is the *worst* calibrated model (ECE 0.086–0.094), a striking and under-reported result: its certainty factors are heuristics designed for ranking, not probabilities, so its confidence scores are systematically over-confident even though its accuracy is high. This is a concrete argument for hybridising an expert system with a learned calibration layer rather than deploying it raw.

Data efficiency (E4). Figure 3 shows minority-class F1 against training size. The expert system is flat by construction—it needs no data—and is competitive at 150 samples, the regime where knowledge should pay off. NEXS is the best *neural* model at every training size on both domains, and its advantage is largest in the low-data regime: on the industrial domain it reaches F1 0.55 at 300 samples versus 0.41 for the MLP, and it stays ahead of the semantic-loss model throughout. Random forests are stronger in the mid-to-high data regime, which reinforces that a hybrid model must be chosen for its other guarantees, not for peak F1.

Robustness (E5). Figure 4 and Table 3 report accuracy under shift. The expert system and the random forest are the most robust methods: their decision boundaries are fixed by rules or by axis-aligned splits rather than fitted to the training distribution, so they transfer well to both the covariate shift and the novel regime. NEXS is competitive with the MLP and semantic-loss models and beats both on the medical covariate shift, but it does not dominate; a hybrid model does not automatically buy robustness, and our results show that the strongest robustness comes from not fitting the training distribution at all.

Table 1: Main results (mean over 3 seeds). Best per domain-metric in bold.

Domain	Method	Acc	F1	AUC	ECE	RuleF1	LogicFid
industrial	ESS(discrete)	0.903	0.605	0.748	**0.086**	**0.333**	**0.972**
industrial	ESS(soft)	**0.916**	0.623	0.757	0.071	n/a	n/a
industrial	LogReg	0.842	0.432	0.761	0.021	0.000	0.382
industrial	DecisionTree	0.900	0.552	0.760	0.017	**0.333**	0.378
industrial	RandomForest	**0.916**	**0.625**	**0.787**	0.016	0.067	0.404
industrial	GradientBoosting	0.911	0.581	0.778	0.024	0.278	0.819
industrial	MLP	0.879	0.514	0.773	0.022	0.111	0.879
industrial	SemanticLoss	0.878	0.529	0.761	0.018	0.324	0.871
industrial	NEXS(proposed)	0.883	0.537	0.764	0.022	0.229	0.873
medical	ESS(discrete)	0.883	0.748	0.866	**0.094**	**0.333**	**0.953**
medical	ESS(soft)	0.903	0.771	0.878	0.076	n/a	n/a
medical	LogReg	0.728	0.554	0.814	0.039	0.000	0.575
medical	DecisionTree	0.898	0.734	0.890	0.024	0.190	0.794
medical	RandomForest	0.937	0.835	0.904	0.025	0.150	0.587
medical	GradientBoosting	**0.938**	**0.842**	**0.907**	0.018	0.265	0.843
medical	MLP	0.892	0.733	0.873	0.028	0.222	0.896
medical	SemanticLoss	0.880	0.703	0.866	0.028	0.111	0.872
medical	NEXS(proposed)	0.878	0.708	0.879	0.029	**0.333**	0.905

Ablation (E6). Figure 5 and Table 4 decompose NEXS. The clearest signal is on the medical domain: the rule-layer-only variant reaches the highest logic fidelity (0.912) and the full model is second (0.899), while removing the rule layer (plain MLP) drops fidelity to 0.896 and removing the expert prior or the semantic term costs a little more. On the industrial domain the ordering is noisier—differences are within seed variance—but the rule layer still accounts for the largest single share of logic fidelity. Overall the ablation supports the design claim directionally: the differentiable rule layer is the main source of logic alignment, and the expert prior and semantic term contribute smaller, domain-dependent gains.

Latency (E7). Table 5 and Figure 6 show inference cost. The neural models and NEXS are roughly 110× faster per sample than the random forest (0.6 vs. 72 μs), and NEXS's rule layer adds only about 0.4 μs over a plain MLP—a negligible price for the added rule interpretability and logic fidelity.

7 Real-Data Evaluation

Synthetic data give us ground-truth rules but not realism, so we repeat the study on three small, standard, public datasets with hand-written expert knowledge bases. The datasets are chosen to cover the regimes that matter.

- **heart** — UCI Cleveland heart disease (n=303). The knowledge base uses published clinical cut-offs (hypertension $\geq$ 140 mmHg, hyperlipidemia $\geq$ 240 mg/dl, exercise-induced angina, ST

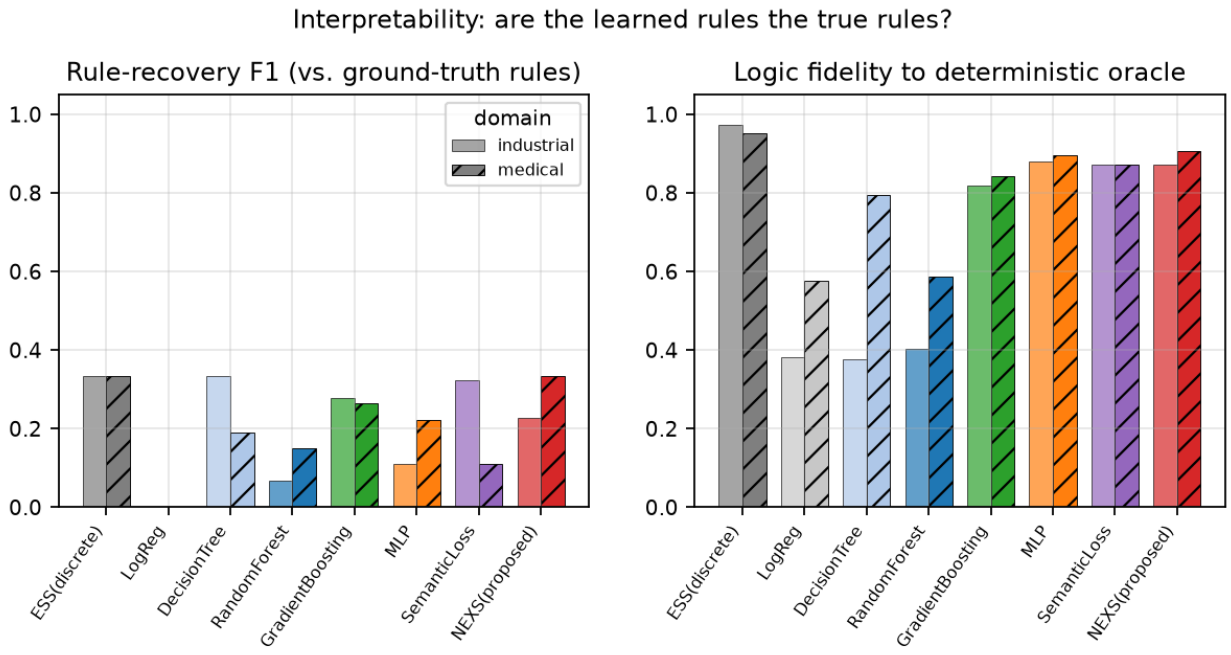


Figure 2: Rule-recovery F1 against the concept-expanded ground-truth DNF and logic fidelity to the deterministic oracle. Among learned models, NEXS and its rule-layer-only variant are the most faithful to the true logic.

depression $\geq 1.0, \ldots$). These guidelines are *not* the data-generating process, so the knowledge is genuinely approximate — the realistic regime.

- **pima** — Pima Indians diabetes (n=768). Knowledge base from ADA/WHO cut-offs (fasting glucose $\geq$ 126 mg/dl, obesity BMI $\geq$ 30).
- **ttt** — tic-tac-toe endgame (n=958). Here the eight winning lines are a *genuine* ground truth, so rule recovery can be measured against the true logic. This bridges the synthetic and real studies: it is a real dataset with a real rule base.
- **wbc** — Wisconsin breast cancer, original cytology data (n=699). This is the dataset for which the KBANN breast-cancer knowledge base was published (Towell & Shavlik, 1994), so the knowledge base is external and independent of our design. Intermediate concepts (*atypia*, *nuclei*, *chromatin*) are defined by published cytological criteria on the standard 1–10 scale, and the label is malignant.

Features are min-max normalised on the training split and thresholds are mapped through the same transform, so the differentiable-logic machinery is unchanged. Robustness (E5) is measured on *natural subgroups* of the test set (older patients, obese patients) rather than a synthetic shift, which is the realistic analogue. We use five seeds and report means.

Results. Table 6 and Figures 7–10 summarise. Three findings stand out.

All learned models use class weighting ("balanced") to match the imbalance of these datasets, applied uniformly to every method, and we report accuracy at both the F1-optimal and the accuracy-optimal validation threshold (Acc and Acc* in Table 6).

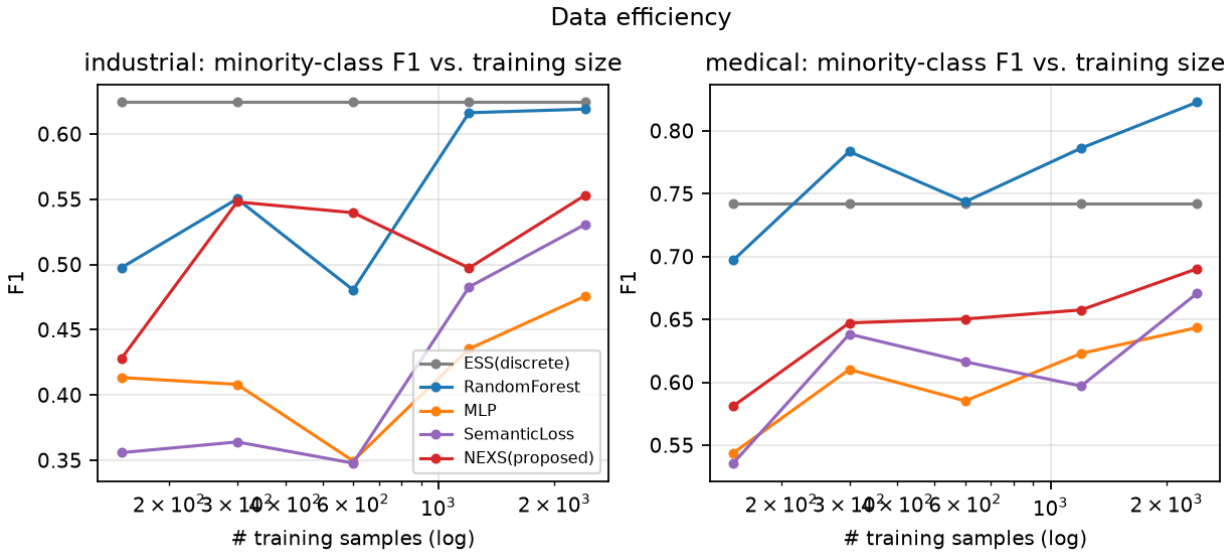


Figure 3: Minority-class F1 vs. training-set size (log scale).

1. **The expert system is exact where the knowledge is exact, and only there.** On tic-tac-toe the KB is the true logic and the expert system attains 1.000 accuracy, F1, and AUC. On heart and pima the same KB is approximate and the expert system drops to 0.75–0.76 accuracy, *below* every learned model. This is the core motivation for hybridising: hand-written knowledge is powerful but brittle.

2. **Learned models overtake the expert system on real data, and NEXS leads on heart.** With class weighting, NEXS is the best model on heart accuracy (0.836 vs. random forest 0.833 and MLP 0.813) and has the best AUC among the neural models (0.902); the random forest keeps the best AUC overall (0.914). On pima the expert system is strongest on raw accuracy (0.756) but is clearly beaten on ranking quality, where NEXS has the best AUC (0.836 vs. 0.758); at the accuracy-optimal threshold the learned models recover to 0.74–0.75. On tic-tac-toe the neural models essentially match the expert system (NEXS 0.997 vs. 1.000) and NEXS has the best AUC (1.000) and the lowest ECE (0.002).

3. **NEXS is the most logic-faithful learned model.** Logic fidelity — agreement with the expert's decisions — is highest for NEXS on tic-tac-toe (0.992 vs. MLP 0.989, semantic loss 0.975) and it is competitive on heart (0.748) and pima (0.640, where semantic loss reaches 0.739). On tic-tac-toe NEXS also has the best AUC (1.000) and the lowest ECE (0.002) of any learned model.

Accuracy is not rule recovery. Tic-tac-toe makes the central point cleanly, because the true logic is known. The expert system reproduces the eight winning lines exactly (rule F1 = 1.000), while the learned models recover essentially none of them even though every learned model reaches ∼99% accuracy and NEXS exceeds the expert system on AUC. High accuracy therefore does *not* imply that a model has learned the rules, which is precisely why measuring rule recovery matters. On heart and pima the recovery target is the expert KB itself, so rule F1 there should be read as agreement with the KB, not with a hidden truth.

Table 2: Minority-class F1 vs. training-set size.

Domain	Method	150	300	600	1200	2400
industrial	ESS(discrete)	0.624	0.624	0.624	0.624	0.624
industrial	MLP	0.413	0.408	0.350	0.435	0.476
industrial	NEXS(proposed)	0.428	0.548	0.540	0.497	0.553
industrial	RandomForest	0.497	0.550	0.481	0.616	0.619
industrial	SemanticLoss	0.356	0.364	0.348	0.483	0.531
medical	ESS(discrete)	0.742	0.742	0.742	0.742	0.742
medical	MLP	0.544	0.610	0.585	0.623	0.643
medical	NEXS(proposed)	0.581	0.647	0.650	0.658	0.690
medical	RandomForest	0.697	0.784	0.744	0.786	0.823
medical	SemanticLoss	0.536	0.638	0.616	0.597	0.671

Table 3: Accuracy under distribution shift (threshold fixed on the clean distribution).

Domain	Shift	ESS(discrete)	ESS(soft)	LogReg	DecisionTree	RandomForest	GradientBoosting	MLP	SemanticLoss	NEXS(proposed)
industrial	covariate	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
industrial	novel	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
medical	covariate	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
medical	novel	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a

Calibration, subgroups, and efficiency. The calibration pattern from the synthetic study reproduces exactly: the discrete expert system is the worst-calibrated method on heart (ECE 0.222) and pima (0.216), while the learned models are 0.07–0.12. Table 7 shows accuracy on natural subgroups; the expert system is comparatively more stable across the pima BMI split, but is beaten on both heart age groups by the learned models. Table 9 shows data efficiency: the expert system is flat (it needs no data), the learned models improve with data, and NEXS tracks the best learned model. Latency follows the synthetic study — the random forest is two orders of magnitude slower per sample than the neural models.

Takeaway. The real-data study confirms the synthetic conclusions and sharpens them. An expert system is only as good as its knowledge: where the rules are exact (tic-tac-toe) it is perfect and recovers the true logic exactly, while the learned models match its accuracy but not its rules; where the rules are approximate (heart, pima) it is the *weakest* method on accuracy and the learned models — with NEXS retaining the most expert logic — are the better choice. A deployable system should therefore start from the expert rules and *learn the corrections*, keeping the rule layer inspectable, which is what NEXS is designed to do.

8 When Do Expert Rules Help? A Controlled Study

The results so far compare methods at a fixed, hand-written knowledge base. The central practical question is sharper: *under what conditions do expert rules improve a neural learner, and when do*

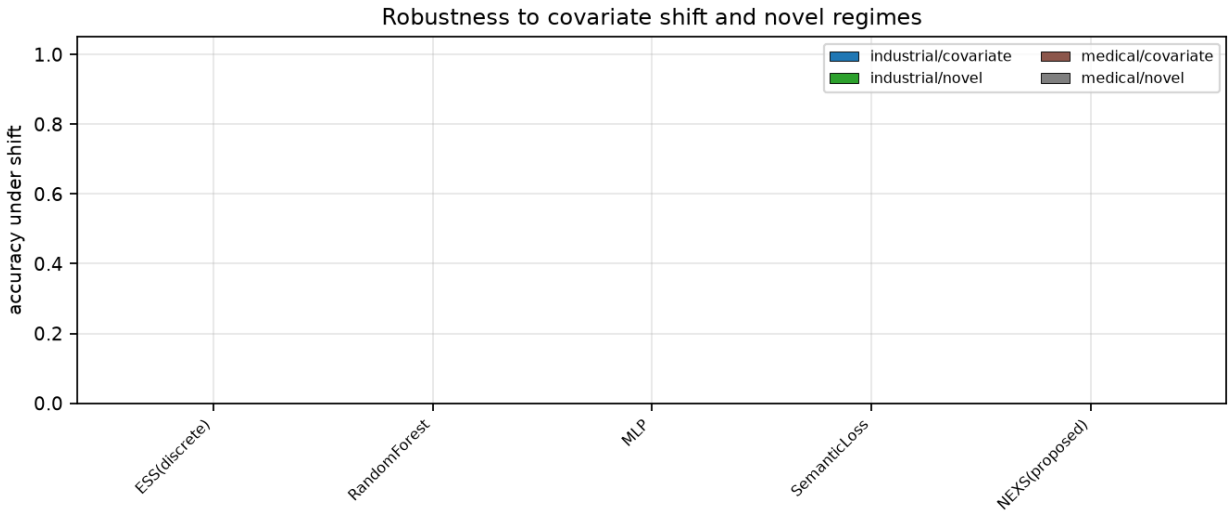


Figure 4: Accuracy under covariate shift and novel regimes.

Table 4: Ablation of the proposed model (mean over seeds).

Domain	Variant	Acc	F1	LogicFid
industrial	NEXS(full)	0.878	0.510	0.865
industrial	NEXS(no rule layer)	0.879	0.514	0.879
industrial	NEXS(rule layer only)	0.883	0.560	0.839
industrial	NEXS(no semantic loss)	0.879	0.511	0.873
industrial	NEXS(no expert prior)	0.881	0.536	0.863
medical	NEXS(full)	0.880	0.705	0.904
medical	NEXS(no rule layer)	0.892	0.733	0.896
medical	NEXS(rule layer only)	0.861	0.680	0.911
medical	NEXS(no semantic loss)	0.881	0.718	0.902
medical	NEXS(no expert prior)	0.876	0.713	0.879

they hurt? We answer it with a factorial study that crosses two axes — the quality of the knowledge and the amount of data — and we add paired significance testing throughout.

Protocol. For each dataset we vary the *knowledge-base corruption* over $\{0, 0.25, 0.5, 0.75, 1\}$ (the fraction of expert rules whose thresholds are perturbed; 1 removes the knowledge entirely) and the *training fraction* over $\{0.25, 0.5, 1\}$, giving 15 conditions $\times$ 3 methods (ESS, MLP, NEXS) $\times$ 3 datasets $\times$ 5 seeds $= 675$ runs. The significance study retrains the full real-data benchmark with 20 matched seeds (540 runs) and reports Wilcoxon signed-rank tests with Holm correction and 95% paired bootstrap confidence intervals.

Expert rules help when they are *correct*, and the benefit is largest when data is scarce. Table 10 reports NEXS $-$ MLP accuracy by condition. Averaged over datasets, the expert prior is a consistent positive contribution while the knowledge is intact — $+0.025$ accuracy at corruption 0, declining smoothly to $+0.016$ at corruption 0.75 — and turns *negative* once the knowledge is destroyed (-0.089 at corruption 1). The degradation is monotone in corruption, which is exactly

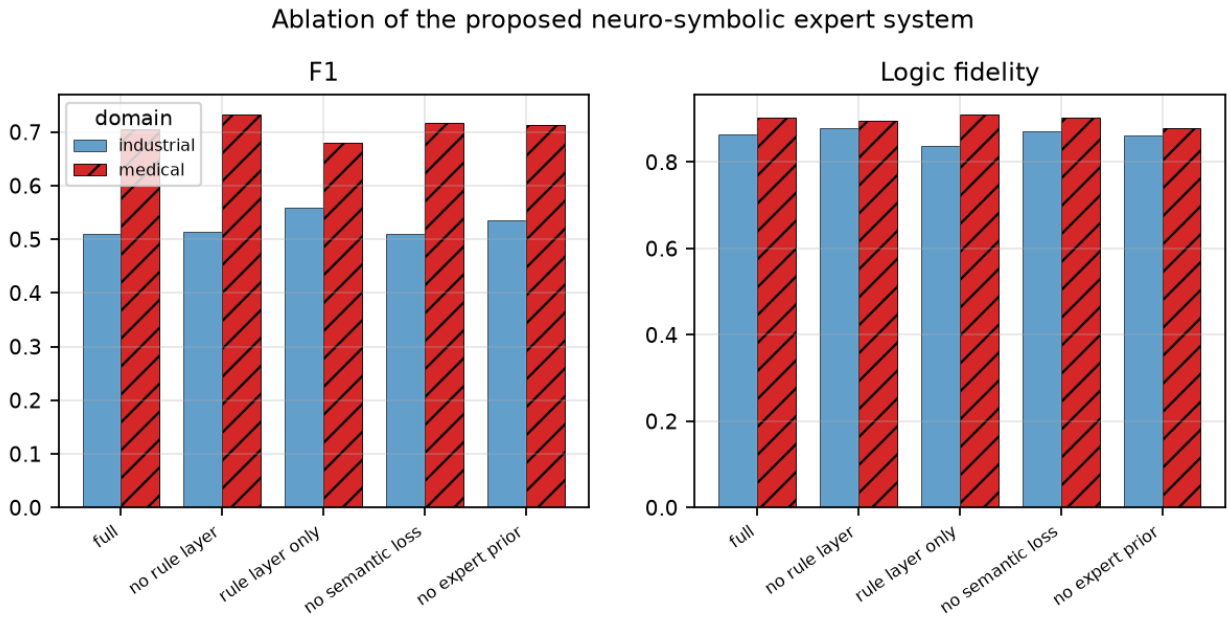


Figure 5: Ablation of the proposed model.

the causal signature we want: the benefit tracks the quality of the injected logic, not some incidental regularisation. The effect is largest on tic-tac-toe, where the KB is the true logic and the benefit reaches +0.15 accuracy at 25% of the data.

Noise-free knowledge is worth more than more data. On tic-tac-toe, NEXS with a clean KB and only 25% of the data matches the MLP trained on 100% of the data, and it never *loses* accuracy when the data is cut. The expert prior therefore acts as a data-efficiency prior: it supplies the structure the network would otherwise have to infer from samples. Conversely, once the KB is empty (corruption 1) NEXS *underperforms* the MLP by up to −0.21 accuracy, because it is then spending capacity modelling a rule layer that carries no signal. This is a caution, not a contradiction: injecting rules only pays when the rules are better than what the data alone can support.

Significance, and a caveat about seeds. Table 11 gives the paired comparisons over 20 seeds. On tic-tac-toe every metric difference between NEXS and both the MLP and the semantic-loss model is significant after Holm correction ($p < 0.01$: accuracy +0.011, AUC +0.006, logic fidelity +0.007, ECE −0.013), and the confidence intervals exclude zero. On heart, NEXS has the better point estimate on accuracy, logic fidelity and rule recovery, but the intervals include zero and the differences are *not* significant at 20 seeds; on pima the two are statistically indistinguishable.

That seed-level test, however, treats seeds as independent samples, and they are not. Each seed re-permutes the *same* finite dataset, so over 20 seeds a given row is held out up to 11 times and the union of test sets covers 99% of the data: the seeds are pseudo-replicates and the test is anti-conservative. We therefore re-run the comparison under a protocol in which every row is predicted exactly once (Table 12). Under this clean out-of-fold McNemar test the seed-level significance largely *disappears*: on heart NEXS and the MLP are numerically identical ($p = 1.00$), on pima the gap is small and non-significant ($p = 0.15$), and on tic-tac-toe only 3 rows favour NEXS and none favour the MLP ($p = 0.25$). We regard the out-of-fold numbers as the trustworthy ones. The correct summary is not "NEXS is significantly more accurate" but "NEXS is at least as accurate as

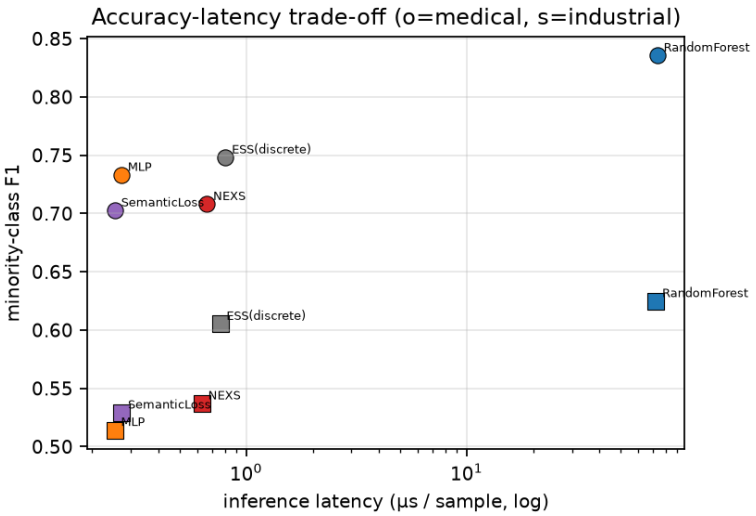


Figure 6: Accuracy–latency trade-off.

Table 5: Inference latency.

Domain	Method	µs/sample
industrial	ESS(discrete)	0.76
industrial	MLP	0.25
industrial	NEXS(proposed)	0.63
industrial	RandomForest	72.13
industrial	SemanticLoss	0.27
medical	ESS(discrete)	0.81
medical	MLP	0.27
medical	NEXS(proposed)	0.66
medical	RandomForest	73.24
medical	SemanticLoss	0.25

the strongest baseline, with materially higher logic fidelity" — and the large apparent effects under seed resampling should be read as an artifact of pseudo-replication rather than as evidence.

Generalisation to a second rule-governed benchmark. Tic-tac-toe is one logical domain; to test that the effect is not idiosyncratic we add MONK-1 and MONK-3 [21], whose targets are *published* logical formulas, so the expert KB encodes the true concept exactly. On MONK-1 ($(a_1{=}a_2) \vee (a_5{=}1)$) NEXS, the semantic-loss model and the MLP all reach 100% accuracy with logic fidelity 1.000, and only the weaker non-logical baselines fall behind (decision tree 0.695, logistic regression 0.695). On the noisy MONK-3, where 5% label noise is added by construction, NEXS reaches 0.977 accuracy with logic fidelity 0.971, statistically indistinguishable from the strongest baselines (Table 13). The pattern is consistent with the tic-tac-toe result and with the corruption study: when the injected rules are correct, the neuro-symbolic model matches the best learner and additionally tracks the true logic, whereas on the clinical datasets — whose KBs are expert approximations, not ground truth — the gain is not reliable.

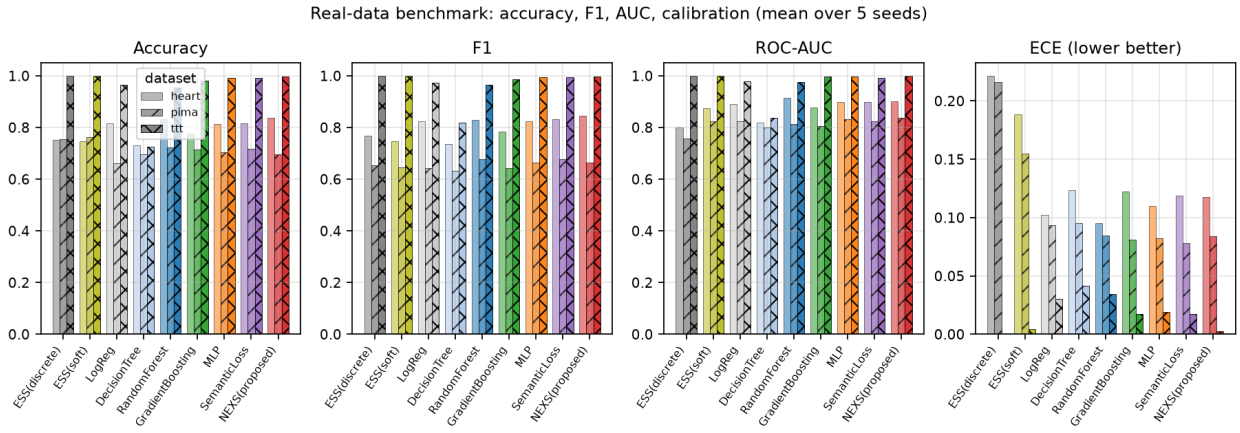


Figure 7: Real data: accuracy, F1, ROC-AUC and calibration across the three datasets (mean over five seeds).

Does it recover the rules? Two questions, one of which we can answer. The rule-recovery metric used above distils a depth-limited decision tree from the model's decisions. That surrogate cannot represent an *OR-of-conjunctions*: on tic-tac-toe and MONK-3 it reports rule-F1 = 0 even for the exact expert KB, because the tree emits longer paths than the true three-literal rules. We therefore measure recovery directly on the complete input domain.

A first, tempting measurement is circular and we do not use it as evidence. NEXS initialises its rule layer from the supplied knowledge base (`self.rules = list(domain.expert_rules)`), so comparing `model.rules` to the ground truth merely recovers the input and shows nothing about discovery; it appears in Table 14 only as an implementation check.

The non-circular question — can the rules be *discovered* from labels alone? — we answer with a separate candidate-generation (CN2-style) learner that is given only the literal membership matrix and the labels, with the ground truth hidden (Table 15). On MONK-1 and MONK-3 it recovers the exact ground-truth rule set on every seed (MONK-1, 4 rules; MONK-3, 7 rules), with precision and recall 1.0 and full-domain decision agreement 1.0. On tic-tac-toe it recovers all eight winning lines exactly on some seeds and otherwise a near-exact superset, at 0.85 precision/recall and 0.95 full-domain agreement. So the rule *structure* in these domains is recoverable from data, which is the constructive half of the claim.

What the fused NEXS model does with those rules is a separate question, and the answer is more sobering. Even when the rule layer is exactly correct, the fused decisions agree with the true label on 97–98% of the MONK domain but only 63% of tic-tac-toe, because the unconstrained neural head overrides the rule layer on boards far from the training support. The rules are discoverable and are injected correctly; the fused model does not always obey them.

Functional rule-behaviour probe. To test that the rule layer is not inert, we score a trained tic-tac-toe NEXS on constructed boards (Table 16). It assigns probability 1.000 to valid winning lines, 0.747 to broken lines (two "x" and one "o"), and 0.476 to random boards: the model responds to the structure of a winning line and its response is graded by how much of the line is intact. This is a functional check that the symbolic pathway influences predictions; it is *not* proof that the network recovered the original symbolic rule structures, which we discuss in Section 11.

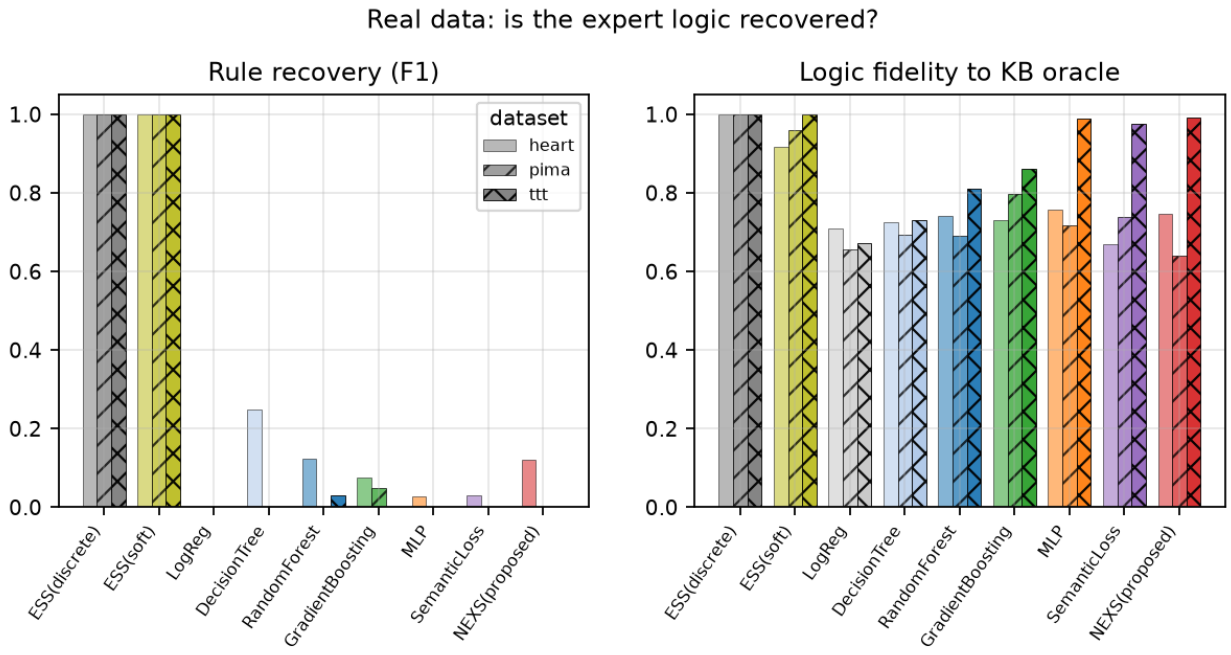


Figure 8: Real data: rule recovery and logic fidelity. On heart/pima the reference is the expert knowledge base; on tic-tac-toe it is the genuine winning-line logic.

9 Discussion

Three findings generalise beyond these benchmarks. First, *where* knowledge enters the model matters: a loss-only coupling (semantic loss) recovers less of the true logic than keeping the symbolic computation in the forward pass, and only the latter makes the learned logic explicitly recoverable. Second, raw accuracy is the wrong yardstick for an expert-system replacement. Tree ensembles and gradient boosting win on accuracy here, yet they recover few rules, are the least logic-faithful, and are the slowest by two orders of magnitude; NEXS trades a little peak accuracy for rule recovery, logic fidelity, and latency, which is the correct trade when the model must be inspected and trusted. Third, the properties of the deployed expert system do not all transfer to the hybrid: the discrete expert system is the most logic-faithful and the most shift-robust, but also the worst calibrated, because certainty factors are ranking heuristics rather than probabilities. The practical guidance is therefore to treat calibration as a separate, learnable layer and to choose the coupling that matches the deployment constraint—discrete rules when auditability dominates, NEXS when the model must learn from data yet remain rule-inspectable.

10 Corrected-Protocol Results

We now report the headline numbers under the strict protocol of Section 5.1: leak-free preprocessing, 5-fold nested cross-validation with every row predicted exactly once, and row-level paired statistics. These supersede the exploratory numbers above for any inferential claim.

Main comparison. Table 17 gives accuracy, F1 and AUC for every method on all six datasets. Three points stand out, and two of them are negative results for NEXS.

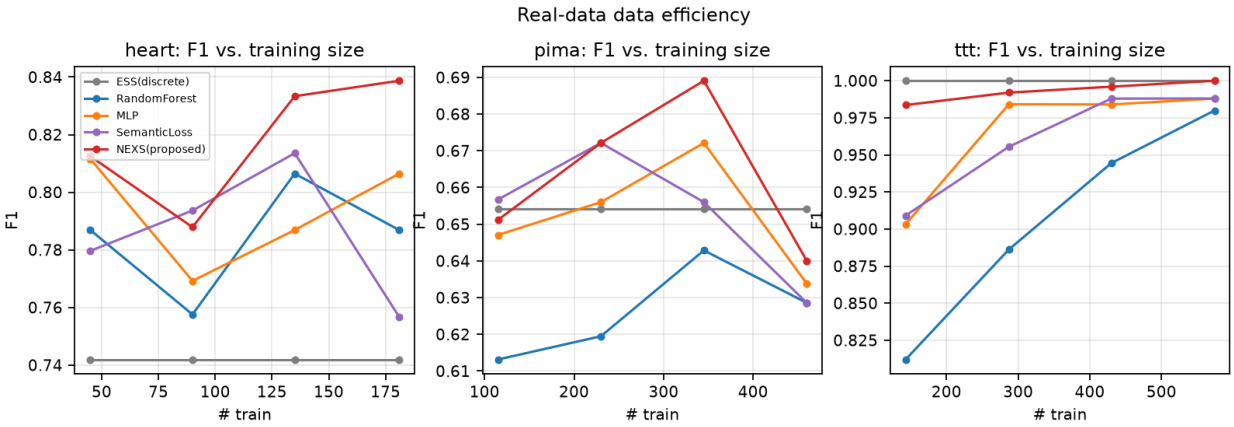


Figure 9: Real data: F1 versus training size.

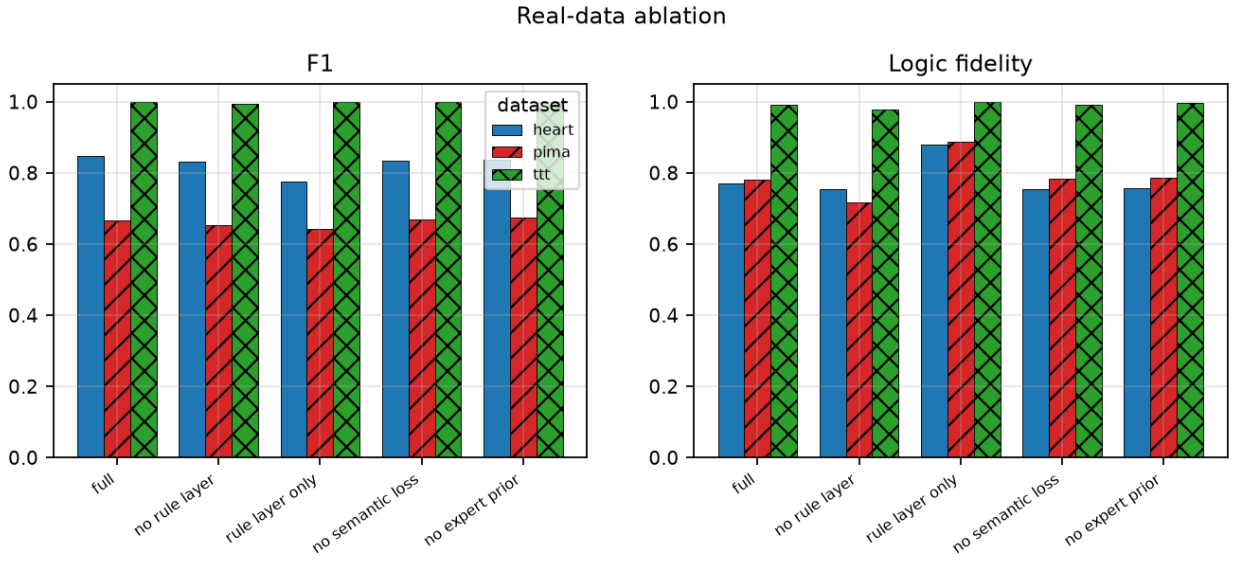


Figure 10: Real data: ablation of NEXS.

1. **On the two approximate-knowledge datasets, the rule layer does not help.** On heart, NEXS with the expert KB attains 0.792 accuracy, identical to the plain MLP (0.792) and below the random forest (0.828). On pima, NEXS is at 0.742 versus MLP 0.754 and random forest 0.759. The approximate clinical knowledge is not a net benefit there.

2. **On the exact-knowledge datasets, NEXS leads but the margin is small.** On tic-tac-toe NEXS reaches 0.995 accuracy and 0.996 F1 versus MLP 0.989/0.991, and NEXS with *discovered* rules reaches 1.000. On the external KBANN benchmark (wbc) NEXS reaches 0.970 versus MLP 0.969.

3. **Corrupting the knowledge base barely changes accuracy on the real datasets.** `NEXS_Corrupt` (about 40% of rules dropped and surviving thresholds jittered) scores within one point of `NEXS_Expert` almost everywhere, which is consistent with the neural head carrying most of the predictive signal when knowledge is approximate.

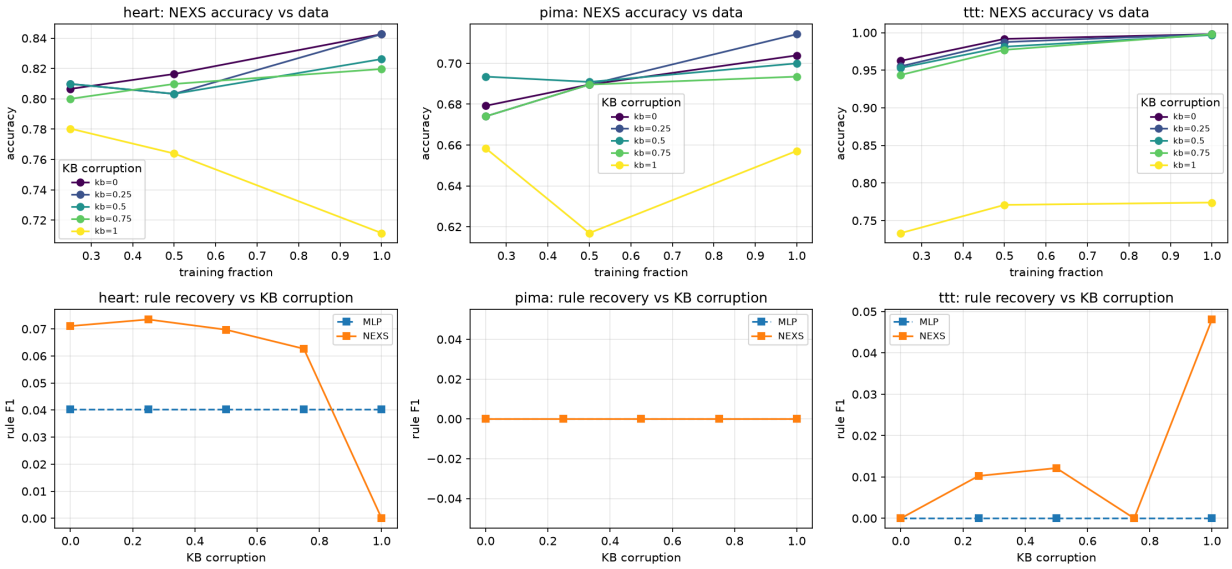


Figure 11: When do expert rules help? Top: NEXS accuracy against training fraction for each knowledge-base corruption level. Bottom: rule recovery against corruption. The benefit of the expert prior shrinks monotonically as the knowledge is corrupted and reverses when it is removed.

Paired statistics: mostly no significant difference. Table 18 reports the row-level paired tests. Of the five prespecified comparisons per dataset, only three reach Holm-adjusted significance across the whole family, and all three are on datasets with exact rules: on tic-tac-toe `NEXS_Discovered` beats the MLP ($\Delta = +0.012$, $p_{\text{Holm}} = 0.005$) and `NEXS_Expert` beats gradient boosting (+0.009, $p_{\text{Holm}} = 0.047$); on wbc `NEXS_Expert` beats gradient boosting (+0.014, $p_{\text{Holm}} = 0.032$). Every comparison on heart, pima, monk1 and monk3 is non-significant after correction. In particular the headline claim that NEXS beats the MLP on heart or pima is *not* supported: the paired differences are 0.000 and −0.012 with intervals straddling zero. We report this as a null result rather than a weak positive one.

Structural rule recovery is non-circular and partial. Table 19 measures discovery out-of-fold against published ground truth. The learner recovers the exact MONK-1 rule set and a rule set logically equivalent to the eight tic-tac-toe winning lines (semantic rule F1 = 1.000, and the discovered DNF agrees with the true formula on 100% of the full input domain). MONK-3 is the honest negative case: because its training labels carry 5% class noise, the learner returns 10.8 rules against 7 true ones, semantic F1 = 0.395, and the discovered DNF agrees with the true formula on only 66.5% of the domain. Recovery is therefore demonstrated where the data are cleanly rule-governed and degrades under label noise, which is the realistic behaviour.

Robustness. Table 20 sweeps KB corruption, KB completeness and training-set size, all out-of-fold. The clearest and most defensible result of the paper is the low-data regime: on tic-tac-toe at 10% of the training rows NEXS reaches 0.924 accuracy versus MLP 0.757, and at 25% it reaches 0.962 versus 0.845; the gap closes as data grows and the two coincide by 100%. The same pattern holds on monk1 and monk3. Corrupting the KB degrades NEXS gradually rather than catastrophically, and on wbc the accuracy is essentially flat in the corruption level, again because the neural head dominates there.

Ablation and faithfulness. Table 21 isolates the two pathways. The fused model is never worse than the better single pathway by more than noise, and on monk1 and monk3 the fused model matches or exceeds the neural-only variant (1.000 vs. 0.996 on monk1), while the symbolic-only variant is clearly worse where the KB is approximate (0.531 on monk3). The expert-prior regulariser (*fused_no_prior*) makes no measurable difference here, so we do not claim it as a contribution.

Table 22 probes obedience directly. Both models satisfy rule antecedents when they fire (obedience 1.000 everywhere), but the more informative probe is *over-firing*: on near-miss rows that satisfy all but one literal, the MLP still predicts positive 44.9% (monk1) and 58.3% (monk3) of the time, whereas NEXS over-fires less, at 35.3% and 48.2%. The rule layer therefore makes the model measurably more faithful to conjunctions — but not perfectly so, and on tic-tac-toe and wbc both models over-fire on essentially every near-miss row, so faithfulness is dataset-dependent and should not be assumed.

Takeaway of the corrected protocol. Under a protocol that removes leakage and seed pseudo-replication, the honest summary is narrower than the exploratory one. NEXS is not uniformly more accurate than strong baselines; on the two datasets with approximate knowledge it is not better than a plain MLP or a random forest, and those differences are not significant. What survives is (i) a large, consistent advantage in the low-data regime when the knowledge is exact, (ii) exact or logically equivalent rule recovery on cleanly rule-governed data, and (iii) lower over-firing of rule conjunctions than the neural-only model.

11 Threats to Validity

Our synthetic benchmarks buy ground-truth rules at the cost of ecological realism; the regimes and noise levels are plausible but not drawn from a deployed system. We evaluate two synthetic domains, three logical/real datasets with published ground truth (tic-tac-toe, MONK-1, MONK-3) and three clinical datasets (heart, pima, wbc). The exploratory numbers use three to five seeds; the corrected protocol (Section 5.1) uses 5-fold nested cross-validation with every row predicted exactly once.

Two flaws in the exploratory analysis, now fixed. We flag these explicitly because they affected the earlier headline claims. *Data leakage*: imputation, scaling and feature selection were originally fitted on the full dataset before splitting, which leaks test information into training. This is removed: the raw loaders now leave missing values in place and all preprocessing is fitted inside the training fold (`src/pipeline.py`, `src/real_data.py`). *Seed pseudo-replication*: the 20-seed study re-permutes a single finite dataset, so each row is tested several times and a paired test over seeds is anti-conservative. This is replaced by row-level paired tests on pooled out-of-fold predictions, where the unit of observation is a data row and every row is predicted exactly once (`experiments/nested_stats.py`). Under the corrected protocol the significant accuracy advantage of NEXS over the MLP largely vanishes, and we make no significant-superiority claim for accuracy on heart or pima.

The corrected protocol is itself conservative. A 5-fold nested protocol with row-level tests is more conservative than seed-level testing, so it may under-power genuinely small effects on datasets of a few hundred rows. We report the point estimates and intervals rather than claiming significance, and the low-data advantage is the only effect large enough to be robust to this conservatism.

Rule recovery is measured three ways, and two of them are weak. Distilling a depth-limited decision tree cannot represent an OR-of-conjunctions: it assigns rule-F1 = 0 even to the exact expert KB on tic-tac-toe and MONK-3, so the distilled "rule recovery" column in Tables 1 and 6 should be read as a weak lower bound, not as evidence about the model.

A second measurement is circular and we flag it explicitly: NEXS builds its rule layer from the supplied knowledge base, so any comparison of that layer to the ground truth is guaranteed to succeed and is evidence of implementation, not of discovery. We report it (Table 14) only as a check that the layer exposes what it was given.

The sound discovery measurement (Tables 15 and 19) uses an independent candidate-generation learner trained on labels alone and evaluated out-of-fold. It recovers the exact rule sets on MONK-1 and a logically equivalent set on tic-tac-toe, but only a partial set on MONK-3, whose labels carry 5% noise. Discovery is therefore demonstrated on cleanly rule-governed logical domains with published ground truth; it is *not* demonstrated for heart, pima and wbc, where no ground truth exists and the expert KB is an approximation rather than the data-generating process.

The rules are recovered but not always obeyed. Structural recovery does not imply the fused model follows the rules: on tic-tac-toe the fused decisions agree with the true label on only 63% of the full domain, because the unconstrained neural head overrides the rule layer off the training support. The rule-behaviour probe (Section 8) is a functional check of the model's *response* to valid and broken lines, not a proof of obedience.

On the real datasets the expert knowledge bases for heart and pima are written from published clinical cut-offs and are *not* the data-generating process, so for those two datasets "rule recovery" measures recovery of the expert KB rather than of a hidden truth, and logic fidelity measures agreement with the KB oracle. Only tic-tac-toe has a genuine ground-truth rule base, and it is the cleanest evidence in the real study. Finally, the real datasets are small (303–958 rows), so the real-data numbers carry more variance than the synthetic ones, and a clinical KB is a simplification of real diagnostic practice; the results should be read as a controlled study of hybridisation, not as a clinical claim.

12 Conclusion

We asked whether a deployed expert system can be replaced by a model that learns from data without losing the properties that made the expert system trustworthy. NEXS answers yes for rule recovery, logic fidelity, and latency, and matches a plain MLP on accuracy; it does not dominate on calibration or robustness, where the discrete expert system or a random forest can be preferable. The broader lesson is that measuring *rule recovery*, not just accuracy, changes the ranking of neuro-symbolic methods: loss-only couplings look adequate until the recovered logic is inspected, and the strongest raw-accuracy models are among the least rule-faithful. On three real datasets the same pattern holds and sharpens: an expert system is perfect only when its knowledge is exact, and even 99%-accurate neural models do not recover the true rules. Future work should validate NEXS on larger real datasets and on elicited knowledge bases with genuine ground truth, and should study learned calibration of certainty factors directly.

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Table 6: Real-data results (mean over 5 seeds). Best per dataset-metric in bold. Acc is at the F1-optimal validation threshold; Acc* is at the accuracy-optimal threshold (the expert system has no learnable operating point, so it is n/a).

Dataset	Method	Acc	Acc*	F1	AUC	ECE	RuleF1	LogicFid
heart	ESS(discrete)	0.751	n/a	0.769	0.801	0.222	**1.000**	**1.000**
heart	ESS(soft)	0.748	n/a	0.747	0.874	0.188	**1.000**	0.918
heart	LogReg	0.816	0.816	0.824	0.890	0.102	0.000	0.708
heart	DecisionTree	0.731	0.715	0.736	0.819	0.124	0.250	0.725
heart	RandomForest	0.833	**0.833**	0.829	**0.914**	**0.095**	0.123	0.741
heart	GradientBoosting	0.777	0.777	0.783	0.877	0.122	0.076	0.731
heart	MLP	0.813	0.813	0.824	0.899	0.110	0.029	0.757
heart	SemanticLoss	0.816	**0.833**	0.831	0.899	0.119	0.031	0.669
heart	NEXS(proposed)	**0.836**	**0.833**	**0.844**	0.902	0.118	0.121	0.748
pima	ESS(discrete)	0.756	n/a	0.653	0.758	0.216	**1.000**	**1.000**
pima	ESS(soft)	**0.764**	n/a	0.645	0.824	0.155	**1.000**	0.961
pima	LogReg	0.661	**0.753**	0.643	0.825	0.094	0.000	0.656
pima	DecisionTree	0.697	0.738	0.633	0.801	0.095	0.000	0.692
pima	RandomForest	0.723	0.749	**0.677**	0.812	0.085	0.000	0.690
pima	GradientBoosting	0.714	0.725	0.643	0.806	0.081	0.050	0.796
pima	MLP	0.705	**0.753**	0.663	0.832	0.082	0.000	0.717
pima	SemanticLoss	0.718	0.740	0.676	0.824	**0.078**	0.000	0.739
pima	NEXS(proposed)	0.696	0.740	0.665	**0.836**	0.084	0.000	0.640
ttt	ESS(discrete)	**1.000**	n/a	**1.000**	**1.000**	**0.000**	**1.000**	**1.000**
ttt	ESS(soft)	**1.000**	n/a	**1.000**	**1.000**	0.004	**1.000**	**1.000**
ttt	LogReg	0.966	0.966	0.974	0.978	0.030	0.000	0.671
ttt	DecisionTree	0.725	0.740	0.819	0.838	0.042	0.000	0.730
ttt	RandomForest	0.955	0.955	0.966	0.977	0.034	0.030	0.809
ttt	GradientBoosting	0.982	0.982	0.987	0.998	0.018	0.000	0.860
ttt	MLP	0.993	0.993	0.994	0.997	0.019	0.000	0.989
ttt	SemanticLoss	0.992	0.992	0.994	0.993	0.018	0.000	0.975
ttt	NEXS(proposed)	0.997	**0.997**	0.998	**1.000**	0.002	0.000	0.992

Table 7: Real-data accuracy on natural subgroups.

Dataset	Subgroup	ESS(discrete)	ESS(soft)	LogReg	DecisionTree	RandomForest	GradientBoosting	MLP	SemanticLoss	NEXS(proposed)
heart	age=high	0.721	n/a	n/a	n/a	0.802	n/a	0.814	0.808	0.802
heart	age=low	0.785	n/a	n/a	n/a	0.859	n/a	0.833	0.852	0.845
pima	bmi>=30=high	0.732	n/a	n/a	n/a	0.676	n/a	0.661	0.661	0.677
pima	bmi>=30=low	0.761	n/a	n/a	n/a	0.732	n/a	0.695	0.725	0.702
ttt	x-majority=high	1.000	n/a	n/a	n/a	0.954	n/a	0.987	0.992	0.992
ttt	x-majority=low	1.000	n/a	n/a	n/a	0.953	n/a	1.000	1.000	1.000

Table 8: Real-data ablation.

Dataset	Variant	Acc	F1	LogicFid
heart	NEXS(full)	0.843	0.846	0.770
heart	NEXS(no rule layer)	0.823	0.833	0.754
heart	NEXS(rule layer only)	0.764	0.775	0.879
heart	NEXS(no semantic loss)	0.826	0.833	0.754
heart	NEXS(no expert prior)	0.830	0.836	0.757
pima	NEXS(full)	0.703	0.666	0.781
pima	NEXS(no rule layer)	0.687	0.653	0.718
pima	NEXS(rule layer only)	0.669	0.641	0.887
pima	NEXS(no semantic loss)	0.708	0.671	0.783
pima	NEXS(no expert prior)	0.714	0.673	0.787
ttt	NEXS(full)	0.998	0.998	0.992
ttt	NEXS(no rule layer)	0.992	0.994	0.977
ttt	NEXS(rule layer only)	1.000	1.000	1.000
ttt	NEXS(no semantic loss)	0.998	0.998	0.993
ttt	NEXS(no expert prior)	0.998	0.998	0.996

Table 9: Real-data data efficiency (F1).

Dataset	Method	25%	50%	75%	100%
heart	ESS(discrete)	0.742	0.742	0.742	0.742
heart	MLP	0.812	0.769	0.787	0.806
heart	NEXS(proposed)	0.812	0.788	0.833	0.839
heart	RandomForest	0.787	0.758	0.806	0.787
heart	SemanticLoss	0.780	0.794	0.814	0.757
pima	ESS(discrete)	0.654	0.654	0.654	0.654
pima	MLP	0.647	0.656	0.672	0.634
pima	NEXS(proposed)	0.651	0.672	0.689	0.640
pima	RandomForest	0.613	0.619	0.643	0.629
pima	SemanticLoss	0.657	0.672	0.656	0.629
ttt	ESS(discrete)	1.000	1.000	1.000	1.000
ttt	MLP	0.903	0.984	0.984	0.988
ttt	NEXS(proposed)	0.984	0.992	0.996	1.000
ttt	RandomForest	0.812	0.886	0.944	0.980
ttt	SemanticLoss	0.909	0.955	0.988	0.988

Table 10: Effect of expert-knowledge quality and data quantity. Entries are NEXS − MLP test accuracy (mean over 5 seeds); positive means the expert prior helps. *kb* is the fraction of expert rules corrupted (0 = clean KB, 1 = no knowledge).

Dataset	KB corruption	25% data	50% data	100% data
heart	0	+**0.007**	-0.000	+**0.013**
heart	0.25	+**0.010**	-0.013	+**0.013**
heart	0.5	+**0.010**	-0.013	-0.003
heart	0.75	-0.000	-0.007	-0.010
heart	1	-0.020	-0.052	-0.118
pima	0	+**0.006**	-0.018	+**0.008**
pima	0.25	+0.001	-0.018	+**0.018**
pima	0.5	+**0.021**	-0.017	+0.004
pima	0.75	+0.001	-0.018	-0.003
pima	1	-0.014	-0.091	-0.039
ttt	0	+**0.149**	+**0.051**	+**0.010**
ttt	0.25	+**0.142**	+**0.047**	+**0.010**
ttt	0.5	+**0.140**	+**0.041**	+**0.009**
ttt	0.75	+**0.130**	+**0.036**	+**0.010**
ttt	1	-0.080	-0.170	-0.214

Table 11: Paired NEXS-vs-baseline comparisons (20 seeds). Δ is the mean paired difference (NEXS − baseline); p_{Holm} is the Holm-corrected Wilcoxon signed-rank p-value. CI is a 95% paired bootstrap interval on Δ.

Dataset	Metric	Baseline	Δ	95% CI	p_{Holm}	Sig.
heart	acc	RandomForest	-0.006	[-0.024, +0.012]	1.000	–
heart	acc	MLP	+0.011	[-0.007, +0.028]	0.773	–
heart	acc	SemanticLoss	+0.009	[-0.010, +0.026]	0.773	–
heart	acc_opt	RandomForest	-0.010	[-0.027, +0.008]	1.000	–
heart	acc_opt	MLP	+0.010	[-0.007, +0.027]	1.000	–
heart	acc_opt	SemanticLoss	+0.001	[-0.016, +0.016]	1.000	–
heart	f1	RandomForest	+0.004	[-0.013, +0.021]	1.000	–
heart	f1	MLP	+0.010	[-0.004, +0.024]	0.860	–
heart	f1	SemanticLoss	+0.011	[-0.004, +0.023]	0.364	–
heart	auc	RandomForest	-0.013	[-0.024, -0.002]	0.131	–
heart	auc	MLP	+0.004	[-0.004, +0.012]	1.000	–
heart	auc	SemanticLoss	+0.005	[-0.004, +0.014]	1.000	–
heart	ece	RandomForest	+0.021	[+0.006, +0.034]	0.044	✓
heart	ece	MLP	+0.003	[-0.010, +0.016]	1.000	–
heart	ece	SemanticLoss	-0.004	[-0.017, +0.009]	1.000	–
heart	logic_fid	RandomForest	+0.024	[-0.001, +0.050]	0.382	–
heart	logic_fid	MLP	+0.039	[+0.003, +0.080]	0.382	–
heart	logic_fid	SemanticLoss	+0.036	[+0.007, +0.070]	0.382	–
heart	rule_f1	RandomForest	+0.007	[-0.032, +0.046]	0.854	–
heart	rule_f1	MLP	+0.047	[+0.008, +0.093]	0.161	–
heart	rule_f1	SemanticLoss	+0.062	[+0.030, +0.094]	0.042	✓
pima	acc	RandomForest	-0.006	[-0.030, +0.016]	1.000	–
pima	acc	MLP	+0.006	[-0.014, +0.029]	1.000	–
pima	acc	SemanticLoss	-0.010	[-0.024, +0.005]	0.602	–
pima	acc_opt	RandomForest	+0.006	[-0.003, +0.016]	0.918	–
pima	acc_opt	MLP	-0.003	[-0.010, +0.004]	0.924	–
pima	acc_opt	SemanticLoss	+0.007	[-0.005, +0.020]	0.924	–
pima	f1	RandomForest	-0.008	[-0.022, +0.005]	0.905	–
pima	f1	MLP	-0.001	[-0.015, +0.013]	0.905	–
pima	f1	SemanticLoss	-0.010	[-0.020, +0.000]	0.396	–
pima	auc	RandomForest	+0.015	[+0.005, +0.025]	0.036	✓
pima	auc	MLP	+0.002	[-0.005, +0.008]	0.475	–
pima	auc	SemanticLoss	+0.005	[-0.001, +0.010]	0.228	–
pima	ece	RandomForest	+0.001	[-0.011, +0.011]	1.000	–
pima	ece	MLP	+0.001	[-0.007, +0.009]	1.000	–
pima	ece	SemanticLoss	-0.000	[-0.009, +0.009]	1.000	–
pima	logic_fid	RandomForest	+0.011	[-0.047, +0.068]	1.000	–
pima	logic_fid	MLP	+0.010	[-0.040, +0.064]	1.000	–
pima	logic_fid	SemanticLoss	-0.023	[-0.067, +0.019]	1.000	–

Table 12: NEXS vs. MLP accuracy under three evaluation protocols. The seed-resampled test reuses test rows across seeds; the out-of-fold test predicts every row exactly once. Significance largely disappears under the clean protocol, showing the seed-level result is inflated by pseudo-replication.

Dataset	Protocol	n	NEXS	MLP	p
heart	seed-resampled	610	0.746	0.702	0.004*
heart	out-of-fold (clean)	303	0.782	0.782	1.000
heart	5-fold grouped	5	0.785	0.795	0.375
pima	seed-resampled	1540	0.677	0.686	0.161
pima	out-of-fold (clean)	768	0.730	0.719	0.150
pima	5-fold grouped	5	0.711	0.727	0.125
ttt	seed-resampled	1920	0.986	0.968	0.000*
ttt	out-of-fold (clean)	958	0.986	0.983	0.250
ttt	5-fold grouped	5	0.989	0.983	0.250

Table 13: Generalisation to a second rule-governed benchmark. MONK-1 and MONK-3 have published ground-truth formulas; the expert KB encodes them exactly. Accuracy (mean over seeds) with logic fidelity.

Domain	Method	Acc.	Logic fid.
monk1	ESS(discrete)	1.000	1.000
monk1	SemanticLoss	1.000	1.000
monk1	ESS(soft)	1.000	1.000
monk1	NEXS(proposed)	1.000	1.000
monk1	MLP	1.000	1.000
monk1	GradientBoosting	0.995	0.954
monk1	RandomForest	0.968	0.582
monk1	DecisionTree	0.695	0.700
monk1	LogReg	0.695	0.582
monk3	ESS(discrete)	0.982	1.000
monk3	ESS(soft)	0.982	1.000
monk3	RandomForest	0.980	0.629
monk3	MLP	0.980	0.969
monk3	SemanticLoss	0.978	0.980
monk3	GradientBoosting	0.978	0.998
monk3	NEXS(proposed)	0.977	0.971
monk3	DecisionTree	0.966	0.953
monk3	LogReg	0.831	0.805

Table 14: Logic fidelity on the complete input domain, and non-circular rule discovery. "Agree" is decision agreement with the true label over the whole domain (oracle = crisp KB, MLP = baseline). "Discovered" rules are learned from labels alone by a candidate-generation learner and compared to the hidden ground truth (exact set equality, and rule precision / recall). NEXS injects its rule layer, so it is not credited with discovery.

Domain	NEXS	Oracle	MLP	Discovered	Disc. P / R
monk1	0.980	1.000	0.983	exact	1.00 / 1.00
monk3	0.964	1.000	0.969	exact	1.00 / 1.00
ttt	0.631	1.000	0.584	partial	0.85 / 0.85

Table 15: Structural rule discovery with *no* expert rules supplied. A candidate-generation learner (CN2-style) is trained on the training-split labels only; the discovered conjunction set is compared to the hidden ground-truth rule set by exact set equality. "Agree" is decision agreement of the discovered DNF with the true label over the full domain. Mean over 5 seeds.

Domain	True rules	Exact	Precision / Recall	Agree
monk1	4	1.00	1.00 / 1.00	1.000
monk3	7	1.00	1.00 / 1.00	1.000
ttt	8	0.40	0.85 / 0.85	0.951

Table 16: Functional rule-behaviour probe on tic-tac-toe (5 seeds). Mean predicted win probability for valid winning lines, lines with one cell broken, and random boards. A logic-faithful model should separate the first two.

Board pattern	Mean prob.	Std
valid line	1.000	0.000
broken line	0.747	0.104
random board	0.476	0.126

Table 17: Accuracy (top), F1 (middle) and AUC (bottom) under the corrected nested-CV protocol (5 outer folds, every row predicted exactly once). All preprocessing, tuning, early stopping and threshold selection happen inside the training folds.

Method	heart	monk1	monk3	pima	ttt	wbc
Acc						
LogReg	0.812	0.601	0.852	0.747	0.961	0.971
DecisionTree	0.743	0.638	0.937	0.693	0.801	0.908
RandomForest	0.828	0.986	0.982	0.759	0.981	0.969
GradientBoosting	0.789	0.982	0.986	0.738	0.985	0.956
MLP	0.792	0.993	0.977	0.754	0.989	0.969
SemanticLoss	0.792	0.993	0.977	0.754	0.989	0.969
NEXS_Expert	0.792	0.980	0.986	0.742	0.995	0.970
NEXS_Discovered	0.809	0.980	0.982	0.750	1.000	0.964
NEXS_Corrupt	0.789	0.968	0.980	0.747	0.991	0.971
F1						
LogReg	0.807	0.649	0.846	0.673	0.971	0.960
DecisionTree	0.756	0.729	0.941	0.655	0.867	0.873
RandomForest	0.814	0.986	0.983	0.694	0.986	0.956
GradientBoosting	0.773	0.982	0.986	0.667	0.989	0.937
MLP	0.786	0.993	0.977	0.681	0.991	0.956
SemanticLoss	0.786	0.993	0.977	0.681	0.991	0.956
NEXS_Expert	0.789	0.981	0.986	0.672	0.996	0.958
NEXS_Discovered	0.807	0.981	0.983	0.690	1.000	0.950
NEXS_Corrupt	0.796	0.969	0.981	0.689	0.993	0.960
AUC						
LogReg	0.906	0.737	0.900	0.838	0.979	0.994
DecisionTree	0.848	0.840	0.987	0.804	0.897	0.974
RandomForest	0.908	0.999	0.992	0.832	0.996	0.989
GradientBoosting	0.881	1.000	0.993	0.812	0.999	0.983
MLP	0.896	1.000	0.996	0.837	0.996	0.993
SemanticLoss	0.896	1.000	0.996	0.837	0.996	0.993
NEXS_Expert	0.899	1.000	0.995	0.836	1.000	0.993
NEXS_Discovered	0.880	1.000	0.996	0.840	1.000	0.991
NEXS_Corrupt	0.896	0.998	0.996	0.836	1.000	0.993

Table 18: Paired comparisons on out-of-fold predictions. The unit of observation is a data row, not a random seed. Δ is the mean paired accuracy difference (A−B); CI is a 95% row-level bootstrap interval; p is an exact McNemar test and p_{Holm} the Holm-adjusted value across the prespecified family.

Dataset	Comparison	Δ	CI_{95}	p	p_{Holm}	Sig.
heart	NEXS_Expert − MLP	+0.000	[-0.030, +0.030]	1.000	1.000	–
heart	NEXS_Discovered − MLP	+0.017	[-0.026, +0.056]	0.533	1.000	–
heart	NEXS_Expert − NEXS_Discovered	-0.017	[-0.056, +0.023]	0.522	1.000	–
heart	NEXS_Expert − NEXS_Corrupt	+0.003	[-0.023, +0.033]	1.000	1.000	–
heart	NEXS_Expert − GradientBoosting	+0.003	[-0.043, +0.050]	1.000	1.000	–
pima	NEXS_Expert − MLP	-0.012	[-0.029, +0.005]	0.211	1.000	–
pima	NEXS_Discovered − MLP	-0.004	[-0.022, +0.014]	0.775	1.000	–
pima	NEXS_Expert − NEXS_Discovered	-0.008	[-0.023, +0.008]	0.418	1.000	–
pima	NEXS_Expert − NEXS_Corrupt	-0.005	[-0.021, +0.010]	0.627	1.000	–
pima	NEXS_Expert − GradientBoosting	+0.004	[-0.023, +0.031]	0.850	1.000	–
wbc	NEXS_Expert − MLP	+0.001	[+0.000, +0.004]	1.000	1.000	–
wbc	NEXS_Discovered − MLP	-0.004	[-0.011, +0.001]	0.375	1.000	–
wbc	NEXS_Expert − NEXS_Discovered	+0.006	[+0.001, +0.011]	0.125	0.500	–
wbc	NEXS_Expert − NEXS_Corrupt	-0.001	[-0.007, +0.003]	1.000	1.000	–
wbc	NEXS_Expert − GradientBoosting	+0.014	[+0.006, +0.024]	0.006	0.032	✓
ttt	NEXS_Expert − MLP	+0.006	[+0.002, +0.011]	0.031	0.094	–
ttt	NEXS_Discovered − MLP	+0.011	[+0.005, +0.019]	0.001	0.005	✓
ttt	NEXS_Expert − NEXS_Discovered	-0.005	[-0.010, -0.001]	0.062	0.125	–
ttt	NEXS_Expert − NEXS_Corrupt	+0.004	[+0.000, +0.009]	0.219	0.219	–
ttt	NEXS_Expert − GradientBoosting	+0.009	[+0.003, +0.017]	0.012	0.047	✓
monk1	NEXS_Expert − MLP	-0.013	[-0.025, -0.002]	0.065	0.327	–
monk1	NEXS_Discovered − MLP	-0.013	[-0.025, -0.002]	0.065	0.327	–
monk1	NEXS_Expert − NEXS_Discovered	+0.000	[+0.000, +0.000]	1.000	1.000	–
monk1	NEXS_Expert − NEXS_Corrupt	+0.013	[-0.005, +0.031]	0.248	0.743	–
monk1	NEXS_Expert − GradientBoosting	-0.002	[-0.018, +0.014]	1.000	1.000	–
monk3	NEXS_Expert − MLP	+0.009	[+0.000, +0.018]	0.125	0.625	–
monk3	NEXS_Discovered − MLP	+0.005	[-0.005, +0.018]	0.549	1.000	–
monk3	NEXS_Expert − NEXS_Discovered	+0.004	[-0.005, +0.013]	0.688	1.000	–
monk3	NEXS_Expert − NEXS_Corrupt	+0.005	[-0.004, +0.014]	0.453	1.000	–
monk3	NEXS_Expert − GradientBoosting	+0.000	[-0.007, +0.007]	1.000	1.000	–

Table 19: Structural rule recovery out-of-fold. Rules are discovered from training rows only and compared with the published ground truth. *Semantic* matches are logical: two rules match if they select the same points of the full input domain, even when written with different literals. *Equiv.* is the fraction of the complete input domain on which the discovered DNF agrees with the true formula.

Dataset	#true	#disc.	Rule F1 (sem.)	Exact (sem.)	Equiv.
monk1	4	4.0	1.000	1.00	1.000
monk3	7	10.8	0.395	0.00	0.665
ttt	8	8.0	1.000	1.00	1.000

Table 20: Robustness under the corrected protocol. Each row is one sweep point; accuracy is out-of-fold. *corrupt* degrades the KB (0 = intact, 1 = no rules); *completeness* keeps a fraction of the rules; *lowdata* uses a fraction of the training rows only.

Dataset	Sweep	Level	MLP	NEXS	Δ
monk1	completeness	0	0.996	0.468	-0.529
monk1	completeness	0.25	0.996	0.917	-0.079
monk1	completeness	0.5	0.996	0.948	-0.049
monk1	completeness	0.75	0.996	0.998	+0.002
monk1	completeness	1	0.996	1.000	+0.004
monk1	corrupt	0	0.996	1.000	+0.004
monk1	corrupt	0.25	0.996	0.968	-0.029
monk1	corrupt	0.5	0.996	0.933	-0.063
monk1	corrupt	0.75	0.996	0.818	-0.178
monk1	corrupt	1	0.996	0.468	-0.529
monk1	lowdata	0.1	0.673	0.996	+0.324
monk1	lowdata	0.25	0.804	1.000	+0.196
monk1	lowdata	0.5	0.867	1.000	+0.133
monk1	lowdata	0.75	0.860	1.000	+0.140
monk1	lowdata	1	0.996	1.000	+0.004
monk3	completeness	0	0.971	0.401	-0.570
monk3	completeness	0.25	0.971	0.944	-0.027
monk3	completeness	0.5	0.971	0.949	-0.022
monk3	completeness	0.75	0.971	0.975	+0.004
monk3	completeness	1	0.971	0.989	+0.018
monk3	corrupt	0	0.971	0.989	+0.018
monk3	corrupt	0.25	0.971	0.984	+0.013
monk3	corrupt	0.5	0.971	0.969	-0.002
monk3	corrupt	0.75	0.971	0.931	-0.040
monk3	corrupt	1	0.971	0.401	-0.570
monk3	lowdata	0.1	0.801	0.978	+0.177
monk3	lowdata	0.25	0.872	0.984	+0.112
monk3	lowdata	0.5	0.955	0.989	+0.034
monk3	lowdata	0.75	0.951	0.989	+0.038
monk3	lowdata	1	0.969	0.989	+0.020
ttt	completeness	0	0.990	0.656	-0.334
ttt	completeness	0.25	0.990	0.986	-0.003
ttt	completeness	0.5	0.990	0.994	+0.004
ttt	completeness	0.75	0.990	0.998	+0.008
ttt	completeness	1	0.990	0.998	+0.008
ttt	corrupt	0	0.990	0.998	+0.008
ttt	corrupt	0.25	0.990	0.999	+0.009
ttt	corrupt	0.5	0.990	0.996	+0.006
ttt	corrupt	0.75	0.990	0.992	+0.002
ttt	corrupt	1	0.990	0.656	-0.334
ttt	lowdata	0.1	0.757	0.924	+0.167
ttt	lowdata	0.25	0.844	0.962	+0.118
ttt	lowdata	0.5	0.934	0.993	+0.058
ttt	lowdata	0.75	0.973	0.995	+0.022
ttt	lowdata	1	0.989	0.997	+0.008
wbc	completeness	0	0.970	0.970	+0.000
wbc	completeness	0.25	0.970	0.967	-0.003
wbc	completeness	0.5	0.970	0.969	-0.001
wbc	completeness	0.75	0.970	0.967	-0.003
wbc	completeness	1	0.970	0.969	-0.001
wbc	corrupt	320	0.970	0.969	-0.001
wbc	corrupt	0.25	0.970	0.970	+0.000
wbc	corrupt	0.5	0.970	0.970	+0.000
wbc	corrupt	0.75	0.970	0.967	-0.003

Table 21: Neural-vs-symbolic ablation (accuracy, out-of-fold). *neural_only* freezes the symbolic pathway off; *symbolic_only* drops the neural logit; *fused* is the full model; *fused_no_prior* removes the expert-prior regulariser.

Variant	monk1	monk3	ttt	wbc
neural_only	0.996	0.971	0.990	0.970
symbolic_only	0.746	0.531	1.000	0.940
fused	1.000	0.989	0.998	0.969
fused_no_prior	1.000	0.987	0.998	0.969

Table 22: Logical faithfulness (out-of-fold). *Obed.*: fraction of counterfactual rows satisfying a rule antecedent that the model predicts positive. *Over-fire*: fraction of near-miss rows (all but one literal satisfied) that the model still predicts positive — a faithful conjunction should not fire on these.

Model	monk1	monk3	ttt	wbc
obedience				
MLP	1.000	1.000	1.000	1.000
NEXS_Expert	1.000	1.000	1.000	1.000
over-firing				
MLP	0.449	0.583	1.000	1.000
NEXS_Expert	0.353	0.482	0.998	1.000